List of Publications

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11 November 2026

A) Submitted Manuscripts / Preprints

1. Ramakanth Madhugiri, Nadja Karl, Devadatta Gosavi, Sarah Krautwurst, Tatjana Koob, Kevin Lamkiewicz, Nora Goldmann, Heiko Slanina, Anna Wacker, Arnold Grünweller, Roland K. Hartmann, **Michael T. Wolfinger**, Manja Marz, Harald Schwalbe, Joanna Sztuba-Solinska, and John Ziebuhr. A conserved alphacoronavirus 3'-UTR RNA element is essential for viral replication. Preprint, 2026

2. Martyna Nowacka, Julita Nowicka, Joanna Grochowska, Magdalena Kulma, Agnieszka Belczyk-Ciesielska, Katarzyna M. Gluchowska, Krzysztofa Odzywol, Agnieszka Zagozdzon, Tomasz K. Wirecki, Chandran Nithin, Bartlomiej Surpeta, Xiaobing Zhang, Jannan Zhao, Lukasz Joachimiak, Bartlomiej Hofman, Joanna Sztuba-Solinska, Katarzyna Drzewicka, Angelika Muchowicz, **Michael T. Wolfinger**, Roman Blaszczyk, Irina Tuszynska, Janusz M. Bujnicki, and Zbigniew Zaslona. Identification of a region within the 5'UTR of Interleukin-2 mRNA offers a promising strategy for drug development for autoimmune disorders. Submitted manuscript, 2026

3. Liam Kerr, Dannielle L. Gemmill, Higor S. Pereira, **Michael T. Wolfinger**, and Trushar R. Patel. "Circle of Life"-Zika virus genomic cyclization is controlled by sequence specificity. Submitted manuscript, 2026

B) Refereed Journal Papers

4. Aamir Aman, Leonhard Sidl, Nitchakan Darai, Peter Wolschann, Thanyada Rungrotmongkol, and **Michael T. Wolfinger**. Computational Analysis of Sequence Editability in the Theophylline RNA Aptamer as a Functional RNA Module. *Int. J. Mol. Sci.*, 27(16):7228, 2026, doi:10.3390/ijms27167228

5. Jule Walter, Leonhard Sidl, Katrin Gutenbrunner, Denis Skibinski, Tim Kolberg, Ivo L. Hofacker, Hua-Ting Yao, Mario Mörl, and **Michael T. Wolfinger**. Rational design of mechanically active RNAs: *de novo* engineering of functional exoribonuclease-resistant RNAs. *Nucleic Acids Res.*, 54(9):gkag473, May 2026, doi:10.1093/nar/gkag473

6. Ivana Borovská, Chundan Zhang, Sarah-Luisa J. Dülk, Edoardo Morandi, Marta F. S. Cardoso, Billal M. Bourkia, Daphne A. L. van den Homberg, **Michael T. Wolfinger**, Willem A. Velema, and Danny Incarnato. Identification of conserved RNA regulatory switches in living cells using RNA secondary structure ensemble mapping and covariation analysis. *Nat. Biotechnol.*, 44(6):989–1001, Jun 2026, doi:10.1038/s41587-025-02739-0. PMID: 40715458

7. Nitchakan Darai, Leonhard Sidl, Thanyada Rungrotmongkol, Peter Wolschann, and **Michael T. Wolfinger**. From structure to function: Computational insights into Musashi-RNA complexes in the context of viral pathogenesis and beyond. *Sci. Asia*, 51S(1):2025s013, Jun 2025, doi:10.2306/scienceasia1513-1874.2025.s013

8. Benoit Besson, Gijs J. Overheul, **Michael T. Wolfinger**, Sandra Junglen, and Ronald P. van Rij. Pan-flavivirus analysis reveals sfRNA-independent, 3'UTR-biased siRNA production from an insect-specific flavivirus. *J. Virol.*, 98(11):e01215–24, Oct 2024, doi:10.1128/jvi.01215-24. PMCID: PMC11575252

9. Lan-Lan Wang, Qia Cheng, Natalee D. Newton, **Michael T. Wolfinger**, Mahali S. Morgan, Andrii Slonchak, Alexander A. Khromykh, Tian-Yin Cheng, and Rhys H. Parry. Xinyang flavivirus, from *Haemaphysalis flava* ticks in Henan province, China, defines a basal, likely tick-only *Orthoflavivirus* clade. *J. Gen. Virol.*, 105(5):001991, May 2024, doi:10.1099/jgv.0.001991. PMCID: PMC11165663

10. Jakob McBroome, Adriano de Bernardi Schneider, Cornelius Roemer, **Michael T. Wolfinger**, Angie S. Hinrichs, Aine Niamh O'Toole, Christopher Ruis, Yatish Turakhia, Andrew Rambaut, and Russell Corbett-Detig. A framework for automated scalable designation of viral pathogen lineages from genomic data. *Nat. Microbiol.*, 9(2):550–560, Feb 2024, doi:10.1038/s41564-023-01587-5. PMCID: PMC10847047

11. Dannielle L. Gemmill, Corey R. Nelson, Maulik D. Badmalia, Higor S. Pereira, Liam Kerr, **Michael T. Wolfinger**, and Trushar R. Patel. The 3' terminal region of Zika virus RNA contains a conserved G-quadruplex and is unfolded by human DDX17. *Biochem. Cell Biol.*, 102(1):96–105, Feb 2024, doi:10.1139/bcb-2023-0036. PMID: 37774422

12. Nitchakan Darai, Kowit Hengphasatporn, Peter Wolschann, **Michael T. Wolfinger**, Yasuteru Shigeta, Thanyada Rungrotmongkol, and Ryuhei Harada. A Structural Refinement Technique for Protein-RNA Complexes Based on a Combination of AI-based Modeling and Flexible Docking: A Study of Musashi-1 Protein. *Bull. Chem. Soc. Jpn.*, 96(7):677–685, Jun 2023, doi:10.1246/bcsj.20230092

13. Tyler Mrozowich, Sean M. Park, Maria Waldl, Amy Henrickson, Scott Tersteeg, Corey R. Nelson, Anneke De Klerk, Borries Demeler, Ivo L. Hofacker, **Michael T. Wolfinger**, and Trushar R. Patel. Investigating RNA-RNA interactions through computational and biophysical analysis. *Nucleic Acids Res.*, 51(9):4588–4601, Mar 2023, doi:10.1093/nar/gkad223. PMCID: PMC10201368

14. Roman Ochsenreiter and **Michael T. Wolfinger**. Strukturierte RNAs in Viren. *BIOspektrum*, 29(2):156–158, Mar 2023, doi:10.1007/s12268-023-1907-x. PMCID: PMC10101536 In German

15. Nitchakan Darai, Panupong Mahalapbutr, Peter Wolschann, Vannajan Sanghiran Lee, **Michael T. Wolfinger**, and Thanyada Rungrotmongkol. Theoretical studies on RNA recognition by Musashi 1 RNA–binding protein. *Sci. Rep.*, 12:12137, Jul 2022, doi:10.1038/s41598-022-16252-w. PMCID: PMC9287312

16. Christoph Flamm, Julia Wielach, **Michael T. Wolfinger**, Stefan Badelt, Ronny Lorenz, and Ivo L. Hofacker. Caveats to deep learning approaches to RNA secondary structure prediction. *Front. Bioinform.*, 2:835422, Jul 2022, doi:10.3389/fbinf.2022.835422. PMCID: PMC9580944

17. Marlena Rozner, Ella Nukarinen, **Michael T. Wolfinger**, Fabian Amman, Wolfram Weckwerth, Udo Bläsi, and Elisabeth Sonnleitner. Rewiring of Gene Expression in *Pseudomonas aeruginosa* During Diauxic Growth Reveals an Indirect Regulation of the MexGHI-OpmD Efflux Pump by Hfq. *Front. Microbiol.*, 13:919539, Jun 2022, doi:10.3389/fmicb.2022.919539. PMCID: PMC9272787

18. Lena S. Kutschera and **Michael T. Wolfinger**. Evolutionary traits of Tick-borne encephalitis virus: Pervasive non-coding RNA structure conservation and molecular epidemiology. *Virus Evol.*, 8(1):veac051, Jun 2022, doi:10.1093/ve/veac051. PMCID: PMC9272599

19. Michael H. D'Souza, Tyler Mrozowich, Maulik D. Badmalia, Mitchell Geeraert, Angela Frederickson, Amy Henrickson, Borries Demeler, **Michael T. Wolfinger**, and Trushar R. Patel. Biophysical Characterisation of Human LincRNA-p21 Sense and Antisense Alu Inverted Repeats. *Nucleic Acids Res.*, 50(10):5881–5898, May 2022, doi:10.1093/nar/gkac414. PMCID: PMC9177966

20. Devadatta Gosavi, Iwona Wower, Irene K. Beckmann, Ivo L. Hofacker, Jacek Wower, **Michael T. Wolfinger**, and Joanna Sztuba-Solinska. Insights into the secondary and tertiary structure of the Bovine Viral Diarrhea Virus Internal Ribosome Entry Site. *RNA Biol.*, 19(1):496–506, Apr 2022, doi:10.1080/15476286.2022.2058818. PMCID: PMC8986297

21. Anastasia Cianciulli Sesso, Branislav Lilić, Fabian Amman, **Michael T. Wolfinger**, Elisabeth Sonnleitner, and Udo Bläsi. Gene Expression Profiling of *Pseudomonas aeruginosa* Upon Exposure to Colistin and Tobramycin. *Front. Microbiol.*, 12:626715, Apr 2021, doi:10.3389/fmicb.2021.626715. PMCID: PMC8120321

22. Hayato Harima, Yasuko Orba, Shiho Torii, Yongjin Qiu, Masahiro Kajihara, Yoshiki Eto, Naoya Matsuta, Bernard M. Hang'ombe, Yuki Eshita, Kentaro Uemura, Keita Matsuno, Michihito Sasaki, Kentaro Yoshii, Ryo Nakao, William W. Hall, Ayato Takada, Takashi Abe, **Michael T. Wolfinger**, Martin Simuunza, and Hirofumi Sawa. An African tick flavivirus forming an independent clade exhibits unique exoribonuclease-resistant RNA structures in the genomic 3'-untranslated region. *Sci. Rep.*, 11(1):4883, Mar 2021, doi:10.1038/s41598-021-84365-9. PMCID: PMC7921595

23. Thomas Spicher, Markus Delitz, Adriano de Bernardi Schneider, and **Michael T. Wolfinger**. Dynamic Molecular Epidemiology Reveals Lineage-Associated Single-Nucleotide Variants That Alter RNA Structure in Chikungunya Virus. *Genes*, 12(2):239, Feb 2021, doi:10.3390/genes120x20239. PMCID: PMC7914560

24. Alexandra Popa, Jakob-Wendelin Genger, Michael D. Nicholson, Thomas Penz, Daniela Schmid, Stephan W Aberle, Benedikt Agerer, Alexander Lercher, Lukas Endler, Henrique Colaco, Mark Smyth, Michael Schuster, Miguel L. Grau, Francisco Martínez-Jiménez, Oriol Pich, Wegene Borena, Erich Pawelka, Zsofia Keszei, Martin Senekowitsch, Jan Laine, Judith H Aberle, Monika Redlberger-Fritz, Mario Karolyi, Alexander Zoufaly, Sabine Maritschnik, Martin Borkovec, Peter Hufnagl, Manfred Nairz, Günter Weiss, **Michael T. Wolfinger**, Dorothee von Laer, Giulio Superti-Furga, Nuria Lopez-Bigas, Elisabeth Puchhammer-Stöckl, Franz Allerberger, Franziska Michor, Christoph Bock, and Andreas Bergthaler. Genomic epidemiology of superspreading events in Austria reveals mutational dynamics and transmission properties of SARS-CoV-2. *Sci. Transl. Med.*, 12(573), Dec 2020, doi:10.1126/scitranslmed.abe2555. PMCID: PMC7857414

25. Christida E. Wastika, Hayato Harima, Michihito Sasaki, Bernard M. Hang'ombe, Yuki Eshita, Yongjin Qiu, William W. Hall, **Michael T. Wolfinger**, Hirofumi Sawa, and Yasuko Orba. Discoveries of Exoribonuclease-Resistant Structures of Insect-Specific Flaviviruses Isolated in Zambia. *Viruses*, 12(9):1017, Sep 2020, doi:10.3390/v12091017. PMCID: PMC7551683

26. Elisabeth Sonnleitner, Petra Pusic, **Michael T. Wolfinger**, and Udo Bläsi. Distinctive regulation of carbapenem susceptibility in *Pseudomonas aeruginosa* by Hfq. *Front. Microbiol.*, 11:1001, May 2020, doi:10.3389/fmicb.2020.01001. PMCID: PMC7264166

27. Adriano de Bernardi Schneider, Roman Ochsenreiter, Reilly Hostager, Ivo L. Hofacker, Daniel Janies, and **Michael T. Wolfinger**. Updated Phylogeny of Chikungunya Virus Suggests Lineage-Specific RNA Architecture. *Viruses*, 11(9):798, Aug 2019, doi:10.3390/v11090798. PMCID: PMC6784101

28. Adriano de Bernardi Schneider and **Michael T. Wolfinger**. Musashi binding elements in Zika and related Flavivirus 3'UTRs: A comparative study *in silico*. *Sci. Rep.*, 9(1):6911, May 2019, doi:10.1038/s41598-019-43390-5. PMCID: PMC6502878

29. Flavia Bassani, Isabelle Anna Zink, Thomas Pribasnig, **Michael T. Wolfinger**, Alice Romagnoli, Armin Resch, Christa Schleper, Udo Bläsi, and Anna La Teana. Indications for a moonlighting function of translation factor aIF5A in the crenarchaeum *Sulfolobus solfataricus*. *RNA Biol.*, 16(5):675–685, May 2019, doi:10.1080/15476286.2019.1582953. PMCID: PMC6546411

30. Roman Ochsenreiter, Ivo L. Hofacker, and **Michael T. Wolfinger**. Functional RNA Structures in the 3'UTR of Tick-borne, Insect-specific and No Known Vector Flaviviruses. *Viruses*, 11(3):298, Mar 2019, doi:10.3390/v11030298. PMCID: PMC6466055

31. Petra Pusic, Elisabeth Sonnleitner, Beatrice Krennmayr, Dorothea Agnes Heitzinger, **Michael T. Wolfinger**, Armin Resch, and Udo Bläsi. Harnessing Metabolic Regulation to increase Hfq-dependent Antibiotic Susceptibility in *Pseudomonas aeruginosa*. *Front. Microbiol.*, 9:2709, Nov 2018, doi:10.3389/fmicb.2018.02709. PMCID: PMC6237836

32. Maria Waldl, Bernhard C. Thiel, Roman Ochsenreiter, Alexander Holzenleiter, João Victor de Araujo Oliveira, Maria Emília M.T. Walter, **Michael T. Wolfinger**, and Peter F. Stadler. TERribly Difficult: Searching for Telomerase RNAs in Saccharomycetes. *Genes*, 9(8):372, Jul 2018, doi:10.3390/genes9080372. PMCID: PMC6115765

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35. Elisabeth Sonnleitner, Alexander Wulf, Sébastien Campagne, Xue-Yuan Pei, **Michael T. Wolfinger**, Giada Forlani, Konstantin Prindl, Laetitia Abdou, Armin Resch, Frederic Allain, Ben Luisi, Henning Urlaub, and Udo Bläsi. Interplay between the catabolite repression control protein Crc, Hfq and RNA in Hfq-dependent translational regulation in *Pseudomonas aeruginosa*. *Nucleic Acids Res.*, 46(3):1470–1485, Feb 2018, doi:10.1093/nar/gkx1245. PMCID: PMC5815094

36. Muralidhar Tata, Fabian Amman, Vinay Pawar, **Michael T. Wolfinger**, Siegfried Weiss, Susanne Häussler, and Udo Bläsi. The anaerobically induced sRNA PaiI affects denitrification in *Pseudomonas aeruginosa* PA14. *Front. Microbiol.*, 8:2312, Nov 2017, doi:10.3389/fmicb.2017.02312. PMCID: PMC5703892

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41. Martin Hölzer, Verena Krähling, Fabian Amman, Emanuel Barth, Stephan H. Bernhart, Victor Carmelo, Maximilian Collatz, Gero Doose, Florian Eggenhofer, Jan Ewald, Jörg Fallmann, Lasse M. Feldhahn, Markus Fricke, Juliane Gebauer, Andreas J. Gruber, Franziska Hufsky, Henrike Indrischek, Sabina Kanton, Jörg Linde, Nelly Mostajo, Roman Ochsenreiter, Konstantin Riege, Lorena Rivarola-Duarte, Abdullah H. Sahyoun, Sita J. Saunders, Stefan E. Seemann, Andrea Tanzer, Bertram Vogel, Stefanie Wehner, **Michael T. Wolfinger**, Rolf Backofen, Jan Gorodkin, Ivo Grosse, Ivo L. Hofacker, Steve Hoffmann, Christoph Kaleta, Peter F. Stadler, Stephan Becker, and Manja Marz. Differential transcriptional responses to Ebola and Marburg virus infection in bat and human cells. *Sci. Rep.*, 6(1):34589, Oct 2016, doi:10.1038/srep34589. PMCID: PMC5054393

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44. Muralidhar Tata, **Michael T. Wolfinger**, Fabian Amman, Nicole Roschanski, Andreas Dötsch, Elisabeth Sonnleitner, Susanne Häussler, and Udo Bläsi. Rnaseq based transcriptional profiling of *Pseudomonas aeruginosa* PA14 after short- and long-term anoxic cultivation in synthetic cystic fibrosis sputum medium. *PLoS ONE*, 11(1):e0147811, Jan 2016, doi:10.1371/journal.pone.0147811. PMCID: PMC4731081

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48. Martin Mann, Marcel Kucharík, Christoph Flamm, and **Michael T. Wolfinger**. Memory efficient RNA energy landscape exploration. *Bioinformatics*, 30(18):2584–2591, Sep 2014, doi:10.1093/bioinformatics/btu337. PMCID: PMC4155248

49. Fabian Amman, **Michael T. Wolfinger**, Ronny Lorenz, Ivo L. Hofacker, Peter F. Stadler, and Sven Findeiß. TSSAR: TSS annotation regime for dRNA-seq data. *BMC Bioinformatics*, 15(1), Mar 2014, doi:10.1186/1471-2105-15-89. PMCID: PMC4098767

50. Ivo L. Hofacker, Christoph Flamm, Christian Heine, **Michael T. Wolfinger**, Gerik Scheuermann, and Peter F. Stadler. BarMap: RNA folding on dynamic energy landscapes. *RNA*, 16(7):1308–1316, Jul 2010, doi:10.1261/rna.2093310. PMCID: PMC2885680

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C) Refereed Conference Papers

55. Dominik Scheuer, Frederic Runge, Jörg K.H. Franke, **Michael T. Wolfinger**, Christoph Flamm, and Frank Hutter. Bayesian approximation of RNA folding times. In *ICLR 2025 Workshop on AI for Nucleic Acids*, Mar 2025. doi:10.5281/zenodo.15228717

56. Dominik Scheuer, Frederic Runge, Jörg K. H. Franke, **Michael T. Wolfinger**, Christoph Flamm, and Frank Hutter. KinPFN: Bayesian approximation of RNA folding kinetics using prior-data fitted networks. In *The Thirteenth International Conference on Learning Representations (ICLR 2025)*, Jan 2025. doi:10.5281/zenodo.15233965

57. Maria Waldl, Sebastian Will, **Michael T. Wolfinger**, Ivo L. Hofacker, and Peter F. Stadler. Bi-alignments as Models of Incongruent Evolution of RNA Sequence and Secondary Structure. In Paolo Cazzaniga, Daniela Besozzi, Ivan Merelli, and Luca Manzoni, editors, *Computational Intelligence Methods for Bioinformatics and Biostatistics*, pages 159–170. Springer International Publishing, Dec 2020. doi:10.1007/978-3-030-63061-4_15

58. Sebastian Pötzsch, Gerik Scheuermann, Peter F. Stadler, **Michael T. Wolfinger**, and Christoph Flamm. Visualization of lattice-based protein folding simulations. In *Tenth International Conference on Information Visualisation (IV'06)*, pages 89–94, Washington, DC, USA, Jul 2006. IEEE Computer Society. doi:10.1109/IV.2006.127

D) Refereed Conference Abstracts

59. Darren L. Gemmill, Higor S. Pereira, Maulik Badmalia, Corey R. Nelson, **Michael T. Wolfinger**, and Trushar R. Patel. Identification and characterisation of G-quadruplexes from viral genomes. *Biophys. J.*, 122(3):444a, Feb 2023, doi:10.1016/j.bpj.2022.11.2395

60. Sean M. Park, Tyler Mrozowich, **Michael T. Wolfinger**, and Trushar R. Patel. Investigating Japanese encephalitis virus long-range terminal region interactions. *Biophys. J.*, 121(3):206A, Feb 2022, doi:10.1016/j.bpj.2021.11.1703

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62. Adriano de Bernardi Schneider and **Michael T. Wolfinger**. The role of arbovirus genome untranslated regions on neurotropism. *Int. J. Infect. Dis.*, 79:142, Feb 2019, doi:10.1016/j.ijid.2018.11.347

E) Book Chapters

63. **Michael T. Wolfinger**, Roman Ochsenreiter, and Ivo L. Hofacker. Functional RNA Structures in the 3'UTR of Mosquito-Borne Flaviviruses. In Dmitrij Frishman and Manja Marz, editors, *Virus Bioinformatics*, pages 65–100. Chapman and Hall/CRC Press, 2021

F) Perspectives and Working Papers

64. **Michael T. Wolfinger**. RNA Structure Before the Experiment: An ensemble view of therapeutic RNA design. RNA Forecast Insights 1, RNA Forecast, September 2026. doi:10.5281/zenodo.22903918. Version 1.1

G) Outreach Articles

65. Adriano de Bernardi Schneider and **Michael T. Wolfinger**. Preventing disease outbreaks with computational biology, how far can we go? *NCT CBNW Newsletter*, 58, Jun 2018, doi:10.5281/zenodo.1463018

H) Software and Data Sets

66. **Michael T. Wolfinger**. RNAConSnake: A workflow for discovering and prioritizing locally conserved RNA secondary structure candidates from multiple sequence alignments. Software, Zenodo, Mar 2026. doi:10.5281/zenodo.19210863. Version 0.2.0

67. Ivana Borovská, Chundan Zhang, Sarah-Luisa J. Dülk, Edoardo Morandi, Marta F. S. Cardoso, Billal M. Bourkia, Daphne A. L. van den Homberg, **Michael T. Wolfinger**, Willem A. Velema, and Danny Incarnato. Data set: Identification of conserved RNA regulatory switches in living cells using RNA secondary structure ensemble mapping and covariation analysis. Data set, Zenodo, Jun 2025. doi:10.5281/zenodo.10357456

68. **Michael T. Wolfinger** and Denise Haslwanter. YFVnext: An interactive Nextstrain resource to study the evolution, genomic diversity, and global spread of Yellow fever virus. Software, Zenodo, Mar 2023. doi:10.5281/zenodo.7785410. Version 1.2

69. **Michael T. Wolfinger**, Adriano de Bernardi Schneider, and Roman Ochsenreiter. CHIKVnext: An interactive Nextstrain resource to study the evolution and global spread of Chikungunya virus. Software, Zenodo, Jan 2023. doi:10.5281/zenodo.5171418. Version 5.1

70. **Michael T. Wolfinger** and Adriano de Bernardi Schneider. VEEnext: An interactive Nextstrain resource to study the evolution and global spread of Venezuelan equine encephalitis serocomplex viruses. Software, Zenodo, Jan 2023. doi:10.5281/zenodo.7524847. Version 2.1

71. Lena S. Kutschera and **Michael T. Wolfinger**. TBEVnext: Molecular epidemiology of Tick-borne encephalitis virus. Data set, ResearchGate, 2021. doi:10.13140/RG.2.2.16177.92003

I) Theses

72. **Michael T. Wolfinger**. *Energy Landscapes of Biopolymers*. PhD thesis, Universität Wien, Oct 2004

73. **Michael T. Wolfinger**. The Energy Landscape of RNA Folding. Master's thesis, Universität Wien, Mar 2001